Frequency tables

1 Judy rolled a die 29 times. She recorded the frequency of each number rolled in the table shown:

Number rolled	Frequency
1	6
2	3
3	0
4	7
5	4
6	9

a Which number was rolled the most?

b Which number did she not roll at all?

2 Monica asked 30 people to choose their favorite type of movie. The results are displayed in the following table:
10 people chose Action, 14 people chose Comedy and 6 people chose Documentary.
What type of movie does each letter represent in the table?

Type of Movie	Frequency
A	10
B	14
C	6

3 Ben asked 35 people about how many siblings they have. He found that 12 people had no siblings, 15 people had one sibling, 3 people had two siblings and 5 people had three siblings.
Create a frequency table showing Ben's results.

4 Mr. Rodriguez recorded the number of pets owned by each of the students in his class. He found that 15 students had no pets, 19 students had one pet, 3 students had two pets and 8 students had three pets.
Create a frequency table showing Mr. Rodriguez's results.

5 In a survey, some people were asked approximately how many minutes they take to decide between brands of a particular product. The results are shown in the frequency table below:

a How many people took part in the survey?

b What proportion of people surveyed took 1 minute to make a decision?

Minutes Taken	Frequency
1	13
2	17
3	12

6 For each of the following scenarios, state [redacted] categories A, B, C and D represent in the corresponding frequency tables:

a Sophia asked 28 of her students to pick their favorite sport. 8 picked Football, 15 picked Volleyball, 5 picked Basketball, and 5 picked Tennis.

Sport	A	B	C	D
Frequency	8	15	5	5

b Rochelle asked her students how they spent their weekend. 6 said Fun Park, 11 said Beach, 10 said Camping and 13 said Skiing.

Activity	A	B	C	D
Frequency	11	13	6	10

c Hannah interviewed 29 people about how they get to school. 13 go by Bicycle, 12 go by Car, 8 go by Bus and 9 go by Train.

Transport	A	B	C	D
Frequency	8	12	9	13

d Mr. Smith asked his 28 students about what they want to be when they grow up. 6 wanted to be a civil engineer, 10 wanted to be a doctor, 12 wanted to be a teacher, and 14 wanted to be a politician.

Job	A	B	C	D
Frequency	10	12	6	14

7 The following table shows the number of trains arriving either on time or late at a particular station:

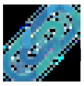
Day	Arriving on time	Arriving late
Monday	23	38
Tuesday	12	25
Wednesday	22	29
Thursday	30	27
Friday	11	14
Saturday	15	11
Sunday	14	10

a How many trains were late on Friday?

b How many trains passed through the station on Wednesday?

c How many trains were on time throughout the entire week?

d What fraction of trains were on time over the whole week?

8 The set of scores for a class of students is  below:

$$81,\ 81,\ 71,\ 81,\ 81,\ 61,\ 71,\ 93,\ 71,\ 58,\ 71,\ 58,\ 61,\ 93,\ 93,\ 71,\ 61,\ 61,\ 81,\ 58,\ 93$$

a Organize the data into a frequency table.

b How many students are there in the class?

c How many students will get a Distinction grade (scores above 80 but less than or equal to 90)?

d How many students will get a High Distinction grade (scores above 90)?

e What is the percentage of students obtaining a High Distinction grade? Round your answer to two decimal places.

9 Below are the luggage weights of 15 passengers, rounded to the nearest kilogram:

$$16,\ \ 19,\ \ 21,\ \ 22,\ \ 19,\ \ 22,\ \ 22,\ \ 17,\ \ 21,\ \ 19,\ \ 16,\ \ 21,\ \ 19,\ \ 16,\ \ 22$$

a Construct a frequency table for this data set.

b How many times did someone check-in luggage that weighed more than 19 kg?

c There is a baggage restriction of 20 kg. The airline charges each passenger $\$1.50$ for every kilogram in their luggage above this restriction. How much was charged altogether for excess baggage?

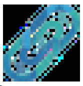
Grouped frequency tables

10 In product testing, the number of faults detected in producing a certain machinery is recorded each day for several days. The results are in the given frequency table:

a Create a column graph to represent the data.

b What is the least possible number of faults that could have been recorded on any particular day?

Number of faults	Frequency
$0-3$	10
$4-7$	14
$8-11$	20
$12-15$	16

11 As part of a fuel watch initiative, the price  gas at a service station was recorded each day for 21 days. The frequency table shows the results:

Price (cents per gallon)	Class Center	Frequency
$130.9 - 135.9$	133.4	6
$135.9 - 140.9$	138.4	5
$140.9 - 145.9$	143.4	5
$145.9 - 150.9$	148.4	5

a What was the greatest price that could have been recorded?

b How many days was the price above 140.9 cents?

12 A survey was conducted which asked 30 people how many books they had read in the past month. Based on the frequency table provided, state whether the following statements are true or false:

a 11 people have read between 6 and 10 books in the past month.

b 28 people have read at most 15 books in the past month.

c We cannot determine from the table how many people have read exactly 12 books.

d We can determine that 2 people have read exactly 5 books in the past month.

Number of books read	Frequency
$1 - 5$	2
$6 - 10$	11
$11 - 15$	15
$16 - 20$	2

13 The following frequency table below shows the resting heart rate of some people taking part in a study:

a Complete the table.

b How many people took part in the study?

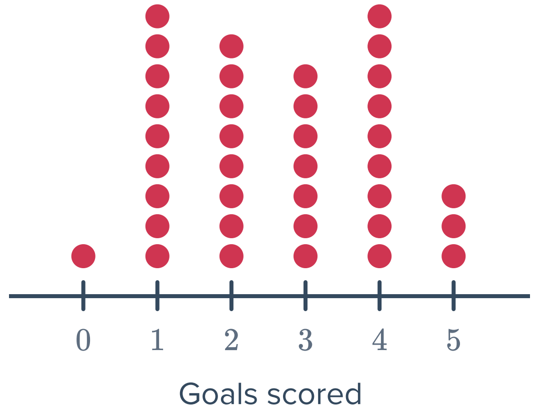
c How many people had a resting heart rate between 55 and 74?

d How many people had a resting heart rate of below 65?

Heart Rate	Class Center	Frequency
$45 - 54$		15
$55 - 64$		25
$65 - 74$		26
$75 - 84$		30

Frequency graphs

14 The goals scored by a football team in their matches are represented in the following dot plot:
Complete a frequency table for this data.

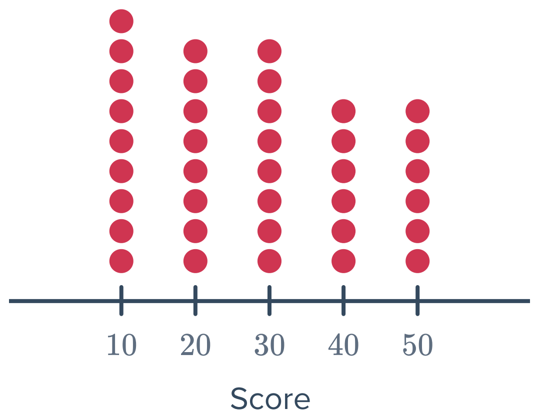


15 Consider the following dot plot showing students scores:

a Complete a frequency table for this data.

b How many students scored above 20?

c How many students scored below 30?



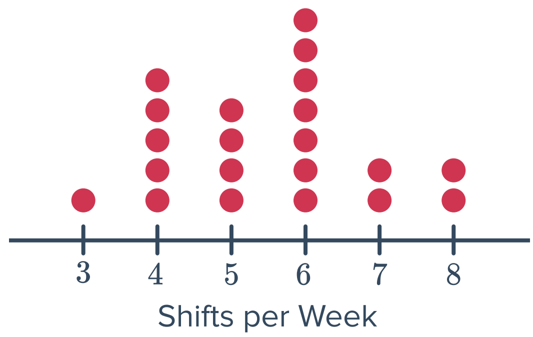
16 Christa is a casual nurse. She used a dot plot to keep track of the number of shifts she did each week for a number of weeks:

a Over how many weeks did Christa record her shifts?

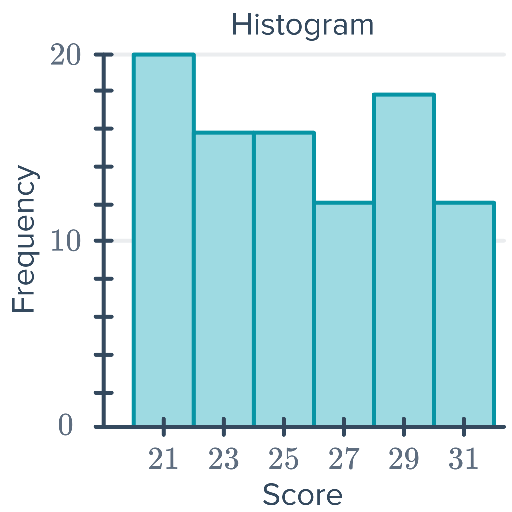
b For how many weeks did she work 5 shifts?

c How many weeks did she work less than 6 shifts?

d If Christa works at least 6 shifts a week, she buys a weekly train ticket. What proportion of the time did she buy a weekly train ticket?



17 Construct a frequency table for the histogram shown:



18 Some people were asked approximately how many of their high school friends they remained in contact with after high school. The results are presented in the given frequency distribution table:

Score	Frequency
0	7
10	10
20	7
30	1
40	2

a Construct a histogram for the data.

b What was the most common response?

19 Construct a histogram to represent the following data:

$$46, \ 47, \ 48, \ 48, \ 48, \ 47, \ 50, \ 47, \ 48, \ 50, \ 47, \ 46, \ 49, \ 49, \ 50, \ 49,$$
$$48, \ 47, \ 46, \ 50, \ 48, \ 51, \ 51, \ 50, \ 50, \ 47, \ 49, \ 48, \ 51, \ 49, \ 50, \ 51$$

20 The amount of snowfall (in centimeters) is recorded at the base of a mountain each day. The results are given below:

$$6, \ 2, \ 0, \ 3, \ 2, \ 2, \ 3, \ 4, \ 2, \ 0, \ 3, \ 2, \ 3, \ 4, \ 6, \ 4, \ 3, \ 0, \ 5, \ 3$$

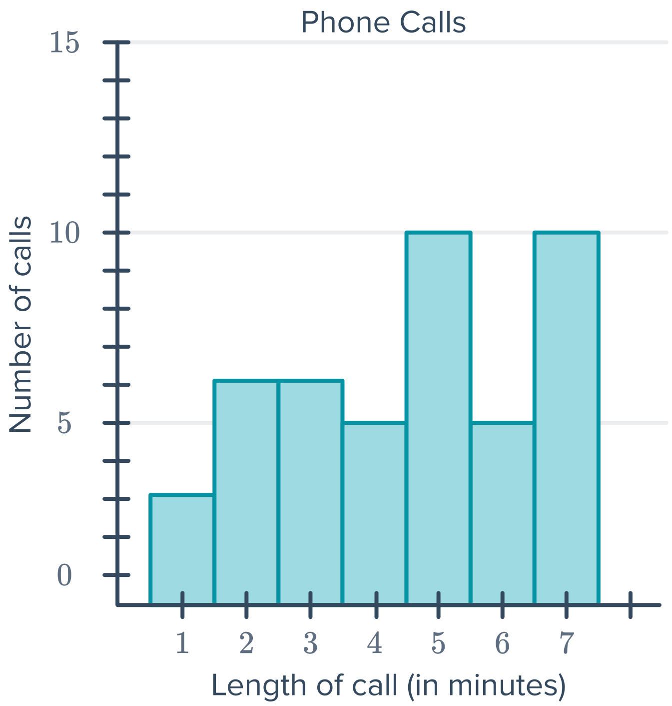
a To create a frequency histogram of the data, which variable goes on the horizontal axis?

b Construct a frequency histogram of the data.

c On how many days did 3 cm of snow fall?

d On how many days did at least 4 cm of snow fall?

21 The histogram below shows the length of [illegible]nber of phone calls made by Daniel:



a Complete a frequency table for the histogram.

b Find the total number of minutes Daniel spent on phone calls.

c If Daniel received 123 minutes of free calls, and was then charged $\$1.00$ per minute for any further calls, how much was he charged for these phone calls?